

ENPH 453 Advanced Physics Laboratory Report 1

Date of submission: February 9, 2026
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Executive Summary

The purpose of this experiment was to measure the mean lifetime of muons. Muons are fundamental subatomic particles that are produced when high-energy charged particles from outer space (cosmic rays) interact with the atmosphere [1]. Despite their short lifespan (~2.2 μs), these muons are detectable at the surface of the Earth due to time dilation [2]. The experiment thus provides direct evidence for special relativistic effects.

A plastic scintillator will emit two flashes of light: first when the muon enters, and second when it decays into an electron/positron and neutrino pair. Negatively charged muons can also be captured by nuclei in the scintillator, shortening their effective lifespan. When coupled to a photomultiplier tube, the scintillation signals can be converted into electronic pulses. Using this with a Time-to-Amplitude Converter (TAC) and various other Nuclear Instrumentation Modules (NIMs), we measured the time between the two events for many muons. The data was collected over 2 weeks and fitted to an exponential decay model, as shown in the figure below.

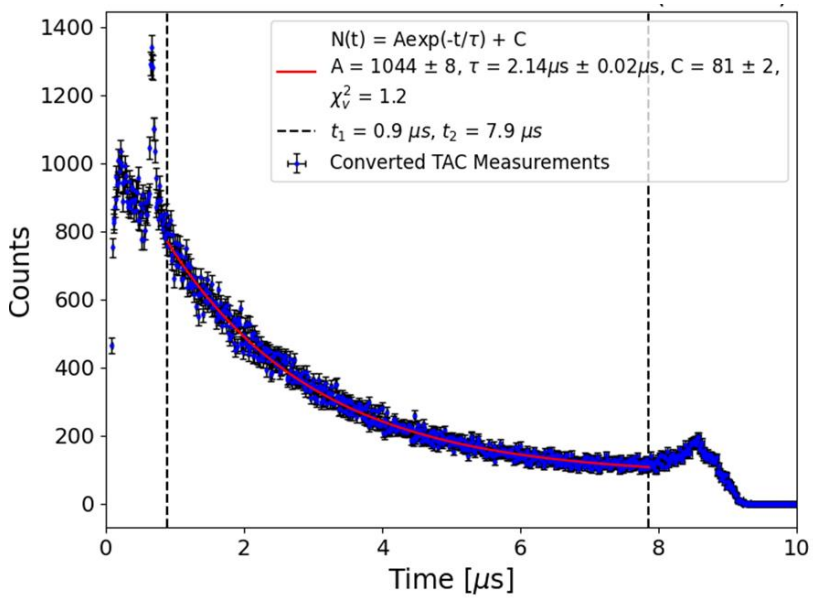


Figure: Plot of counts (y-axis) versus time (x-axis, μs) for a 2-week long muon lifetime measurement. An exponential decay curve was fitted between 0.9μs and 7.9μs. Error bars are also plotted, derived from Poisson statistics and error propagation.

This yielded a mean muon lifetime of $\tau_{meas} = 2.14 \pm 0.02$ μs, which is within uncertainty of the expected value of 2.121 ± 0.008 μs. A double-exponential decay model was also fitted using a fixed flux ratio and negative muon lifetime, leading to a positive muon lifetime of $\tau_{\mu^+} = 2.22 \pm 0.04$ μs. This is within uncertainty of the accepted value of 2.1969811 ± 0.0000022 μs [3]. However, due to statistical uncertainties and the imposed parameter constraints, we are unable to claim that the positive and negative contributions are distinguishable. A repetition of the experiment with a longer data collection period and more careful consideration of systematic errors within the TAC and other NIMs would be necessary to resolve these issues.

Abstract

The purpose of this experiment was to determine the muon lifetime by measuring the time between stopping and decay events produced by cosmic ray muons. This measurement has important implications in particle physics and provides direct experimental evidence for relativistic time dilation. A plastic scintillator coupled to a photomultiplier tube was used to convert scintillation light from muon events into electronic signals, which were processed using Nuclear Instrumentation Modules to isolate muon decays. Data was collected over a two-week period and fit with an exponential decay model, yielding a mean muon lifetime of $\tau_{meas} = 2.14 \pm 0.02$ μs, within uncertainty of the expected value of 2.121 ± 0.008 μs. To account for negative muon capture, a double exponential decay model was also fitted, resulting in an extracted positive muon lifetime of $\tau_{\mu^+} = 2.22 \pm 0.04$ μs, within uncertainty of the accepted value of 2.1969811 ± 0.0000022 μs [3]. However, statistical limitations and strong parameter constraints render this result inconclusive, highlighting the need for a longer data collection period.

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1 Introduction

Muons are fundamental subatomic particles that are a part of the lepton group, possessing a heavier mass and shorter lifespan (~2.2 μs) than the electron [1]. When high-energy charged particles from outer space (cosmic rays) collide with particles in the atmosphere, they produce pions that subsequently decay into muons [1]. These muons travel toward the Earth's surface at relativistic speeds (v ≈ 0.98c) and are detected at sea level with a flux of ~1 $cm^{-2}min^{-1}$ [2].

The observation of muons at the Earth's surface is only possible due to relativistic time dilation. The proper lifetime τ, measured in the muon's rest frame, is related to the lifetime T measured in the Earth's frame with Eq. 1

$$T = \gamma\tau \tag{1}$$

where γ is the Lorentz factor. This allows muons to survive long enough to reach the ground before decaying. If special relativity did not hold, the maximum distance d a muon could travel would simply be the product of its speed v and its lifetime, shown by Eq. 2.

$$d = v\tau = 0.98c \cdot 2.2\mu s = 646m \tag{2}$$

However, the altitude at which muons are produced is ~15 km, far greater than this non-relativistic travel distance [4]. The observation of a substantial muon flux at the Earth's surface thus provides strong experimental evidence for relativistic time dilation and the validity of special relativity. In addition, measuring the muon lifetime allows for the determination of the Fermi coupling constant G_F, which governs the strength of the weak interaction force, and subsequently the mass of the W boson M_W [5]. Therefore, measuring the muon lifetime is important as it has key implications for both particle physics and relativity.

2 Theory

Muons are unstable and will rapidly decay into an electron (or positron) and two neutrinos [6]. The process is mediated by the weak nuclear force, shown below in Figure 1 [7].

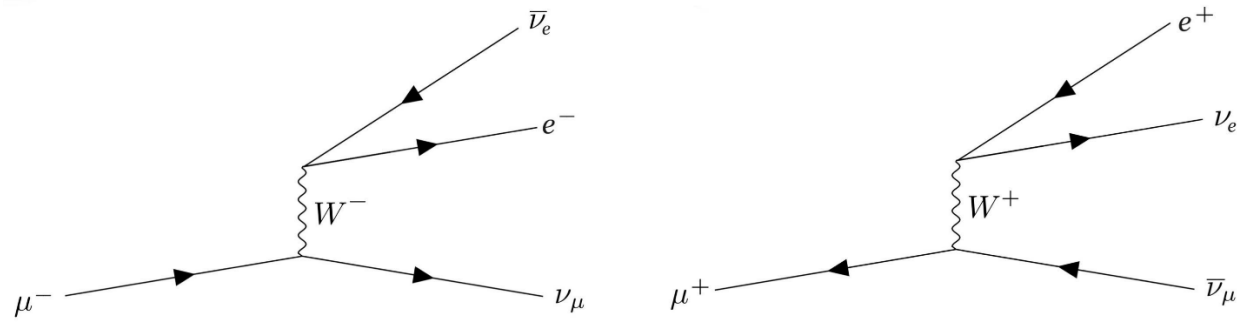


Figure 1: Feynman diagram showing the decay process of a negative muon μ^- (left) and positive muon μ^+ (right), resulting in two neutrinos and an electron/positron.

When a muon slows down inside a scintillator, the deposited energy will generate a brief flash of light [7]. If it also stops, the energy released by the decay process will emit a second flash of light. These events will appear as two sharp, coincident scintillation signals separated by a short delay. By measuring the time between these events for many muons, it is expected that a graph of counts N versus time t will produce an exponential decay curve, given below by Eq. 3

$$N = N_0 e^{-\frac{t}{\tau}} \tag{3}$$

where N_0 is the number of particles at $t = 0$ and τ is the particles lifetime. By fitting this equation to the collected data, it is possible to extract the mean muon lifetime [8]. In addition, since muons are approximately 200 times heavier than the electron [1], a negatively charged muon will orbit extremely close to nearby atomic nuclei. This means there is a high probability for the nuclei to capture the muon, forming a secondary decay channel and resulting in a shorter effective lifespan [7]. This process is shown below in Figure 2.

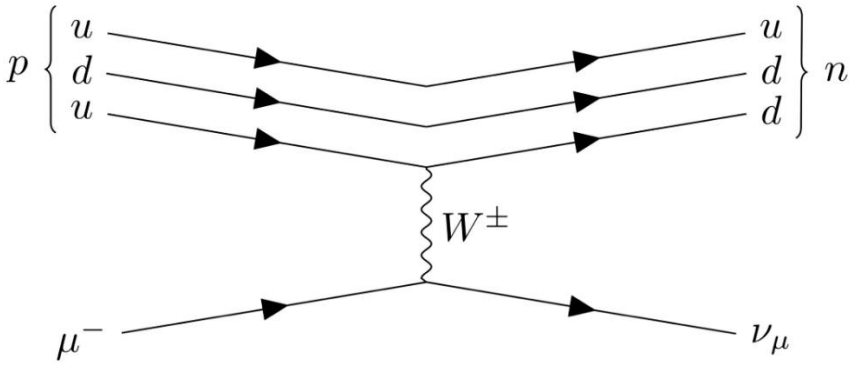


Figure 2: Feynman diagram showing a negative muon μ^- being captured by a proton and subsequently decaying into a neutrino and neutron.

The capture rate Λ_c for a muon is given by the Primakoff Formula [2] (Eq. 4)

$$\Lambda_c(A,Z) = Z_{eff}^4 X_1 \left(1 - X_2\left(\frac{A-Z}{2A}\right)\right) \tag{4}$$

where A and Z are the atomic mass and number respectively for the target material, Z_{eff} is the effective charge the muon experiences, and X_1, X_2 are constants obtained from fits to experimental data [2]. Using this, we can relate the capture rate to the mean negative muon lifetime τ_{μ^-} by using Eq. 5 [9]

$$\tau_{\mu^-} = \left(\Lambda_c + \frac{Q}{\tau_\mu}\right)^{-1} \tag{5}$$

where $0 < Q \leq 1$ is the Huff factor which accounts for the muon decay rate being reduced by its atomic binding energy, and τ_μ is the lifetime of a freely decaying muon. The NE102 scintillator used in this experiment is composed of PVT plastic, which is primarily hydrogen and carbon. Since the capture rate is proportional to Z^4, we can assume that the contribution from hydrogen will be negligible. Using $Z_{eff} = 5.72$ [10], $X_1 = 170 \pm 0.05\ s^{-1}$, $X_2 = 3.125 \pm 0.1$ [2], $A = 12$, and $Z = 6$ for carbon in Eq. 4:

$$\Lambda_c(12,6) = (5.72)^4 \cdot (170 \pm 0.05\ s^{-1}) \cdot \left(1 - (3.125 \pm 0.1) \cdot \left(\frac{12-6}{2 \cdot 12}\right)\right)$$

$$\Lambda_c = 0.039 \pm 0.004\ \mu s^{-1}$$

Using $\Lambda_c = 0.039 \pm 0.004\ \mu s^{-1}$, $Q = 1.00$ [10] and $\tau_\mu = 2.1969811 \pm 0.0000022\ \boldsymbol{\mu s}$ [3] in Eq. 5:

$$\tau_{\mu^-} = \left(0.039 \pm 0.004\ \mu s^{-1} + \frac{1.00}{2.1969811 \pm 0.0000022\ \mu s}\right)^{-1}$$

$$\tau_{\mu^-} = 2.02 \pm 0.02\ \mu s$$

Therefore, we expect negatively charged muons to have a mean lifetime of 2.02 ± 0.02 μs in the plastic scintillator. With a sufficiently large dataset, it is possible to distinguish between the contributions of positively and negatively charged muons [5]. The resulting curve will be a superposition of two decaying exponentials, given below by Eq. 6

$$N = N_0^+ e^{-\frac{t}{\tau_{\mu^+}}} + N_0^- e^{-\frac{t}{\tau_{\mu^-}}} \tag{6}$$

where $\tau_{\mu^+} = \tau_\mu$ since positive muons always decay freely. The expected lifetime τ_{mean} is given by the weighted average between negative and positive muon lifetimes, shown below by Eq. 7

$$\tau_{mean} = \frac{w_+\tau_+ + w_-\tau_-}{w_+ + w_-} \tag{7}$$

where w_+ and w_- are given by the ratio of positive to negative muons. A previous measurement at CERN found that the ratio of positive to negative atmospheric muons at Earth's surface is $\frac{w_+}{w_-} = 1.2766 \pm 0.0032$ [11]. Using this in Eq. 7 with the respective muon lifetimes:

$$\tau_{mean} = \frac{(1.2766 \pm 0.0032)(2.1969811 \pm 0.0000022\ \mu s) + (2.02 \pm 0.02\ \mu s)}{(1.2766 \pm 0.0032) + 1}$$

$$\tau_{mean} = 2.121 \pm 0.008\ \mu s$$

Therefore, the expected mean lifetime is 2.121 ± 0.008 μs.

3 Apparatus and Procedure

3.1 Apparatus

A comprehensive breakdown of the complete apparatus and modules used in the experiment is given below. All components included in the breakdown are pictured in Figure 3.

3.1.1 Oscilloscope

The Tektronix TDS 2012-US digital oscilloscope converts incoming voltage signals into a digital format before displaying the corresponding waveform [12]. It was used throughout the experiment to observe and troubleshoot output pulses, including monitoring the raw scintillation and NIM signals.

3.1.2 Plastic Scintillator

The NE102 plastic scintillator is a large cylinder with a diameter of ~41 cm and a height of ~42 cm. Compared to the scale at which muons are produced (~15 km above the Earth), the scintillator is very small [4]. Thus, muons reaching the detector arrive with zenith angles extremely close to vertical (~0.002^{o}), and the contribution from the sides is negligible [13]. Therefore, the estimated event rate from muons entering the scintillator is (Eq. 8)

$$R = \Phi A = (1\ \mathrm{cm^{-2}min^{-1}})\pi r^2 = (1\ \mathrm{cm^{-2}min^{-1}})(1320\ \mathrm{cm^2}) \approx 1320\ \mathrm{min^{-1}} \qquad (8)$$

However, the measured count rate will also include the decay and capture of low energy muons which fully stop in the scintillator, given by muons with energy less than (Eq. 9)

$$E_{stop} = \left(\frac{dE}{dx}\right)_{plastic} L \approx (2\ \mathrm{MeV \cdot cm^2 \cdot g^{-1}})(1.0\ \mathrm{g \cdot cm^{-3}})(42\ \mathrm{cm}) = 84\mathrm{MeV} \qquad (9)$$

which is only around 1.5% of muons, or an additional 20 min^{-1} [14]. The scintillator has a time constant of 2.4 ns, which is orders of magnitude shorter than the muon lifetime (~2200 ns) [15]. Therefore, the detector response is effectively instantaneous on the timescale of muon decay, allowing coincident muon decay events to be accurately resolved.

3.1.3 Photomultiplier Tube (PMT)

As a muon slows down and decays inside the scintillator, these events produce flashes of light that are detected by the photomultiplier tube (PMT). The PMT converts photons in the UV to visible range into electric pulses [16], [17]. It is made up of a non-thermionic vacuum or glass tube that is comprised of a window for the light to enter, a semitransparent photocathode, an electron-optical input, electron multiplier, and an anode grid [16]. The RC time constant of the

PMT is 0.5ns, which is significantly quicker than the response of the scintillator. This means that the time interval between scintillation events is equivalent to the time between PMT pulses.

3.1.4 Nuclear Instrumentation Modules (NIMs)

The Ortec Model 463 Constant Fraction Discriminator (CFD) converts input pulses into fixed-amplitude logic signals with precise timing [18]. A voltage threshold can be set so that only input pulses above a certain amplitude are transmitted, significantly reducing background noise.

The Ortec Model 449 Log/Lin Ratemeter measures the average count rate for a set range of input pulses [19]. It was used to tune the discriminator by observing the rate of events.

The Ortec Model 566 Time-to-Amplitude Converter (TAC) converts the time between START and STOP pulses into an output pulse with amplitude proportional to the time [20]. This is done by charging a capacitor of capacitance C with a constant current I over the time interval Δt, producing an output voltage V as described by Eq. 10 below.

$$V = \frac{I\Delta t}{C} \tag{10}$$

The Ortec Model 416A Gate and Delay Generator is used to delay input pulses and then output pulses that are identical apart from its polarity [21]. It was used to induce a delay in the successful TAC that was used as the trigger.

The Ortec Model 462 Time Calibrator produces pulses at fixed time intervals and rates [22]. It was used to calibrate the TAC and obtain a conversion factor between time and voltage.

The UG LABS USB-6210 Multi-Channel Analyzer (MCA) converts analog voltage signals into digital outputs, which were collected using the provided LabVIEW program.

The Ortec Model 556 High-Voltage Power Supply produces a steady supply of voltage between 0 to 3 kV, which was used to power the PMT.

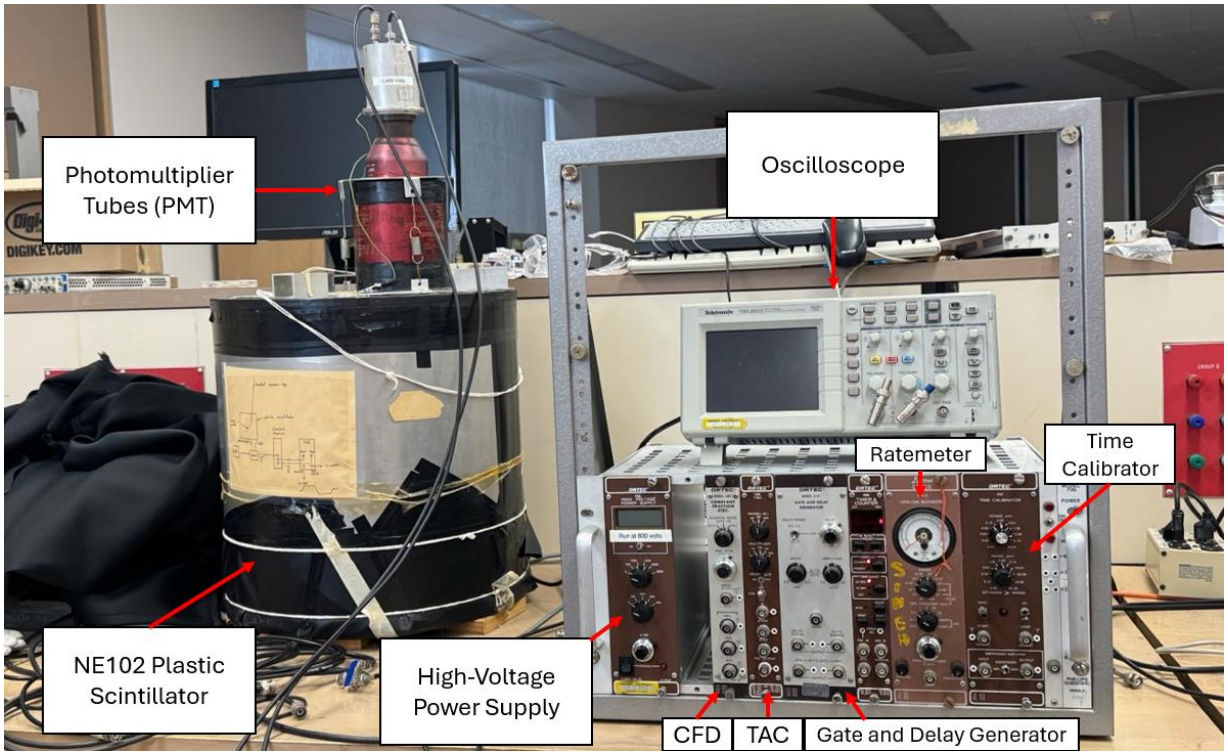


Figure 3: Labelled image of apparatus for the muon lifetime experiment. The abbreviated nuclear instruments stand for Constant Fraction Discriminator (CFD) and Time-to-Amplitude Converter (TAC) respectively.

The connections for the apparatus are shown below in Figure 4.

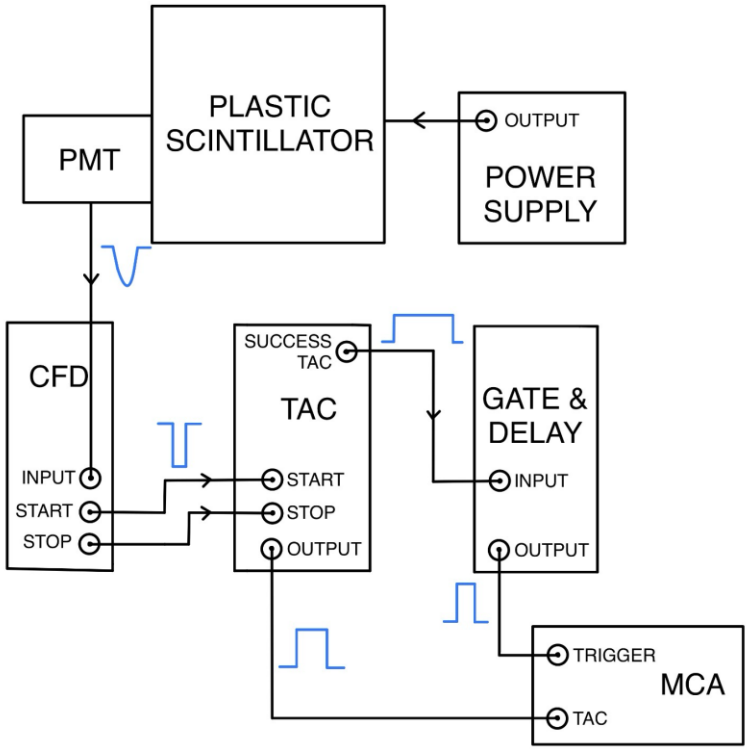


Figure 4: Diagram of connections between Plastic Scintillator, PMT, and NIMs. Blue lines represent the approximate shape of the output signals from each component.

3.2 Procedure

The experimental procedure is outlined in the following steps:

1. The input of the PMT was connected to the high-voltage power supply and to the oscilloscope. The voltage supply was turned on, and its voltage was gradually increased until a significant number of pulses were observed on the oscilloscope (2400V).
2. The output of the PMT was connected to the input of the CFD. The CFD was set to the scintillation mode and the positive output of the CFD was connected to the ratemeter. The threshold of the CFD was slowly increased until the count rate plateaued (900mV). This occurred when the count rate did not noticeably decrease under a small increase in the threshold.
3. The range of the time-to-amplitude converter was set to well above the expected lifetime of a muon (10,000 ns). Next, the START and STOP outputs from the time calibrator were connected into the corresponding inputs on the TAC. The time calibrator was then turned on, and the TAC output signal was confirmed using the oscilloscope. This was done to verify its functionality and confirm that the signal appeared as a square wave with a sharp, linear increase at the beginning.
4. With the time calibrator still connected to the TAC, the successful conversion output located on the rear panel of the TAC was connected to the gate and delay input. Next, the TAC output and gate and delay generator output were both connected into the oscilloscope. The trigger mode on the oscilloscope was set to “normal” and “rising edge” to have it trigger on the gate and delay generator output. The gate and delay were adjusted until the successful conversion signal overlapped with the flat region of the TAC output to ensure that it is after the rising edge. Finally, the gate width was adjusted so that the successful conversion signal was wide enough without exceeding the length of the TAC output signal.
5. The output from the TAC was connected to the MCA input port and the gate and delay output was connected to the MCA trigger input. On the LabVIEW program, the bin count was set to 1000, and the voltage range was set to between 0 to 2 V. The data was recorded for six different time calibrator settings where the rate and interval were set equal to each other. Each configuration was run until a well-defined Gaussian peak formed. Next, the data was exported and a linear fit was performed on the data. By plotting time interval as a function of peak voltage value, the conversion factor was found between voltage and time.
6. The two negative CFD outputs were connected into the START and STOP inputs of the TAC. The gate and delay calibration process from step 4 was repeated for these scintillation pulses.

7. The TAC output was connected to the MCA input, and the gate and delay output was connected to the MCA trigger input. On the LabView program, the bin count was set to 1000, and the voltage range was set between 0 to 10 V. The data collection was started. A correct setup was confirmed when the data slowly formed an exponential decay curve over the span of a few hours and a preliminary mean muon lifetime was extracted.

4 Data Collection and Analysis

This section outlines the data collection and analysis for the TAC calibration and muon lifetime measurements, including curve fitting and preliminary error analysis. The raw data files for all measurements are provided in the Appendix.

4.1 TAC Calibration

The TAC calibration was performed using the instrument settings listed below in Table 1.

Table 1: NIM settings for TAC calibration measurements, performed using a time calibrator.

Instrument	Setting(s)
Time-to-Amplitude Converter	10,000 ns range
Time Calibrator	Range = time intervals between 0.08 µs to 2.56 µs
Gate and Delay Generator	780 ns delay, 800 ns width

The LabVIEW program was run at a 250 kHz frequency, 1000 bins, and a voltage range between 0 to 2 V. Each time interval was run until the peak value reached ~500 counts. The collected data is plotted below in Figure 5 and tabulated in the Appendix.

The error in each bin σ_N is given by Poisson statistics (Eq. 11) where N is the number of counts.

$$\sigma_N = \sqrt{N} \tag{11}$$

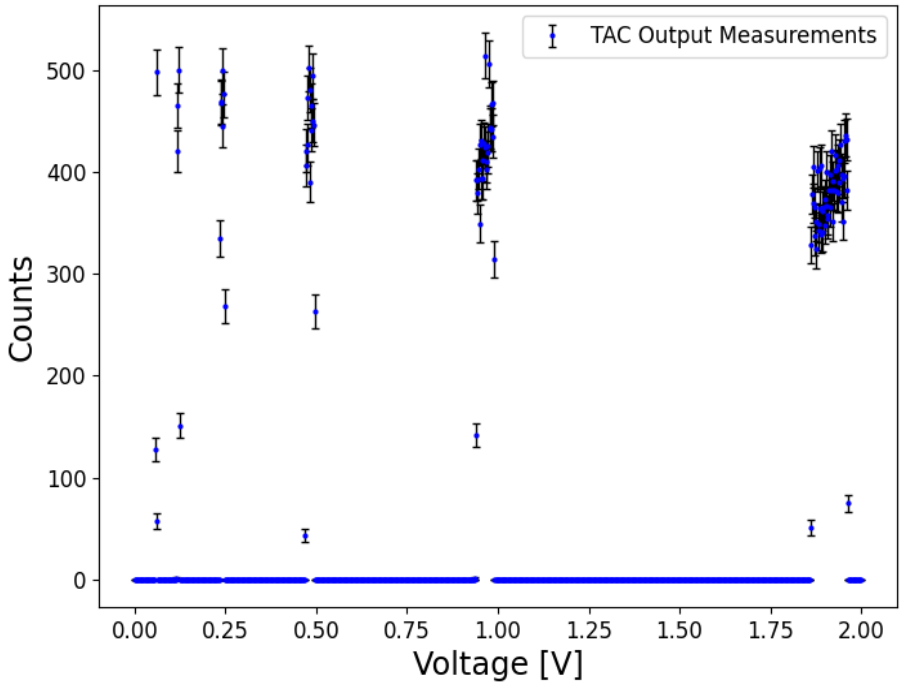


Figure 5: Data obtained from TAC calibration measurements using a time calibrator set at six different time intervals (0.08µs to 2.56µs) and a gate delayed trigger. Counts are plotted against voltage. Vertical error bars assume Poisson statistics ($\sigma = \sqrt{N}$).

Using the Gaussian analyzer in the LabVIEW program, the full width at half maximum (FWHM), peak voltage value, and total counts of each Gaussian curve were also recorded. The last Gaussian curve, which corresponds to a time interval of 2.56 µs, was removed from analysis due to an ill-defined peak. Since the TAC output is linear in voltage V and time t, a conversion factor between the two quantities is obtained by fitting a straight line which follows Eq. 12

$$V = mt + b \tag{12}$$

where m is the slope and b is the intercept. Each measurement produced a normal distribution, so the peak voltage error σ_V is given by the mean error of a Gaussian, shown by Eq. 13

$$\sigma_V = \frac{\sigma}{\sqrt{N}} \tag{13}$$

where N is the total number of counts (area) in each Gaussian curve, and σ is the standard deviation given by Eq. 14.

$$\sigma = \frac{FWHM}{\sqrt{2ln2}} \tag{14}$$

The error in the time interval σ_t was set equal to ±0.005%, quoted directly from the time calibrator specifications sheet [22]. A linear regression using the total least squares method was performed, and the results are shown below in Figure 6.

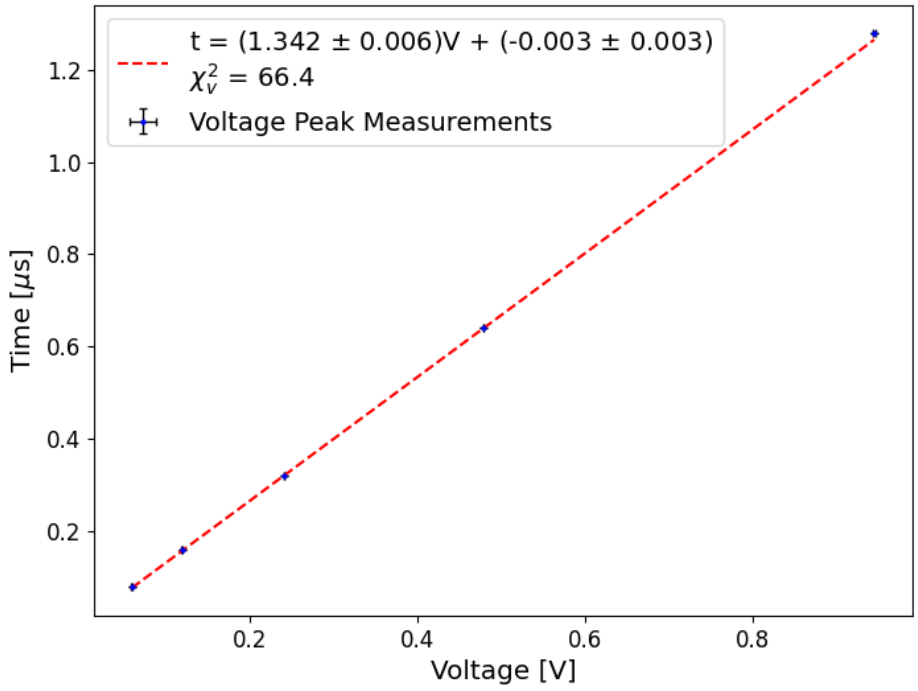


Figure 6: TAC calibration data plotting the peak voltages (V) corresponding to each time calibrator interval (µs). Data was fitted with $V = mt + b$ using the total least squares method. Errors in the x and y directions are plotted but are too small to be discernible.

Table 2 below contains the results of the linear regression in Figure 6.

Table 2: Results of linear regression for the TAC calibration.

Parameter	Value	Error (±)
m	1.342	0.006
b	-0.003	0.003
χ_v^2	66.4	N/A

The high reduced chi squared value of $\chi_v^2 = 66.4$ is caused by the small error in both voltage and time. The waveforms output from the time calibrator must be processed by both the TAC and MCA, likely leading to a higher time interval error than expected. Additionally, it is evident that for smaller voltages the bin widths are too short to specify a well-defined FWHM. This can be seen in Figure 6. Despite this, the data shows a linear trend as expected, so the equation was used in later analysis.

4.2 Mean Muon Lifetime Measurement

A 2-week long data collection of muon decays was performed using the instrument settings shown below in Table 3.

Table 3: NIM settings which were held constant during the muon lifetime measurement.

Instrument	Setting(s)
High-Voltage Power Supply	2400 V
Time-to-Amplitude Converter	10,000 ns range
Constant Fraction Discriminator	900 mV threshold
Gate and Delay Generator	780 ns delay
	800 ns and 400 ns width

Two separate, week long runs were performed with an 800 ns and 400 ns gate width respectively. The LabVIEW program was run with a 250kHz frequency, 1000 bins, and a voltage range between 0 to 8 V. The maximum voltage was chosen after a preliminary 2 daylong run showed that the TAC output ended around ~7 V. The combined data is plotted below in Figure 7 and provided in the Appendix. Poisson statistics were assumed for the y-error (Eq. 11).

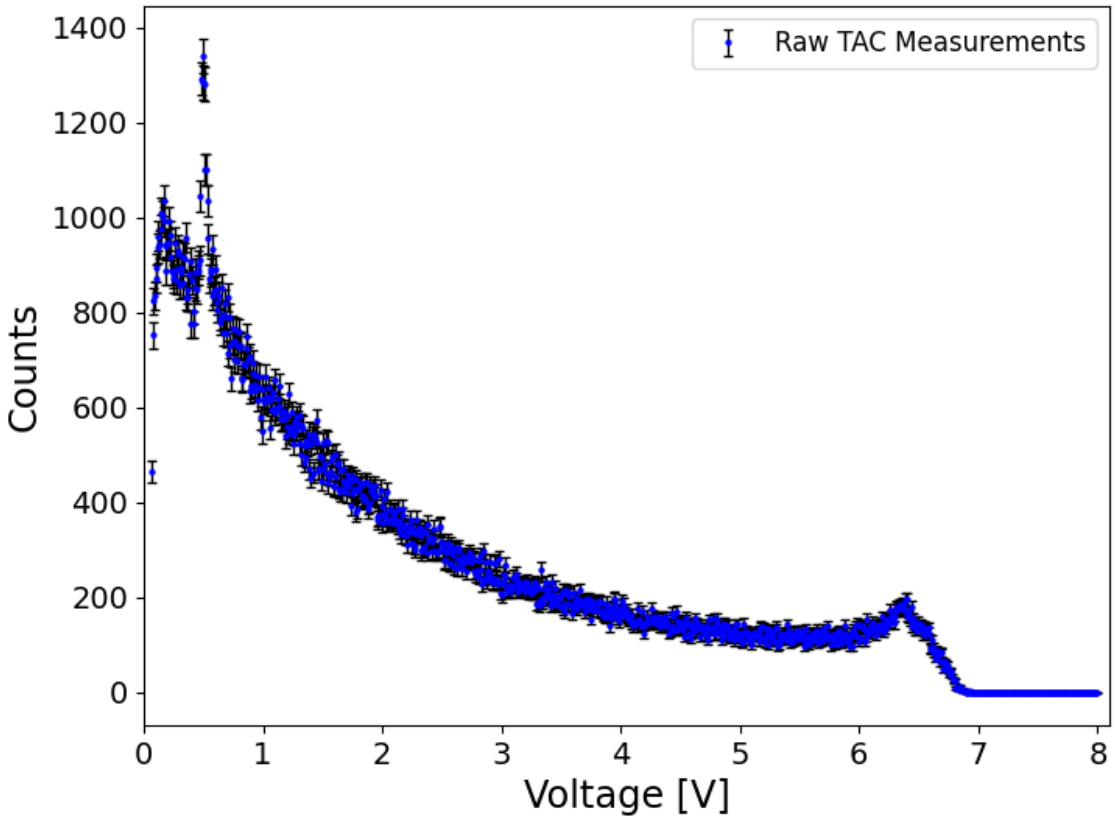


Figure 7: Data obtained from an MCA for the muon lifetime measurement, where a scintillator and PMT setup is used. The measurement is read from a TAC, which converts the decay time into a voltage signal. Plotted are the counts (y-axis) versus voltage (x-axis, V) for a 2-week long measurement, with 1000 bins and a voltage range between 0 to 8V. Error bars in the y-direction are plotted assuming Poisson statistics ($\sigma = \sqrt{N}$).

The voltages were converted into time using Eq. 12, and error was propagated using the standard derivative method. Weighted curve fitting was performed using SciPy's optimization package and an exponential decay formula, given below by Eq. 15

$$N(t) = Ae^{-\frac{t}{\tau}} + C \tag{15}$$

where A, τ, and C are fitting parameters. Erroneous regions, including the sharp peak at ~0.8V and the smooth bump at ~6.5V, were excluded from the fit. Justifications for this are discussed in Section 5. The resulting fit is shown below in Figure 8.

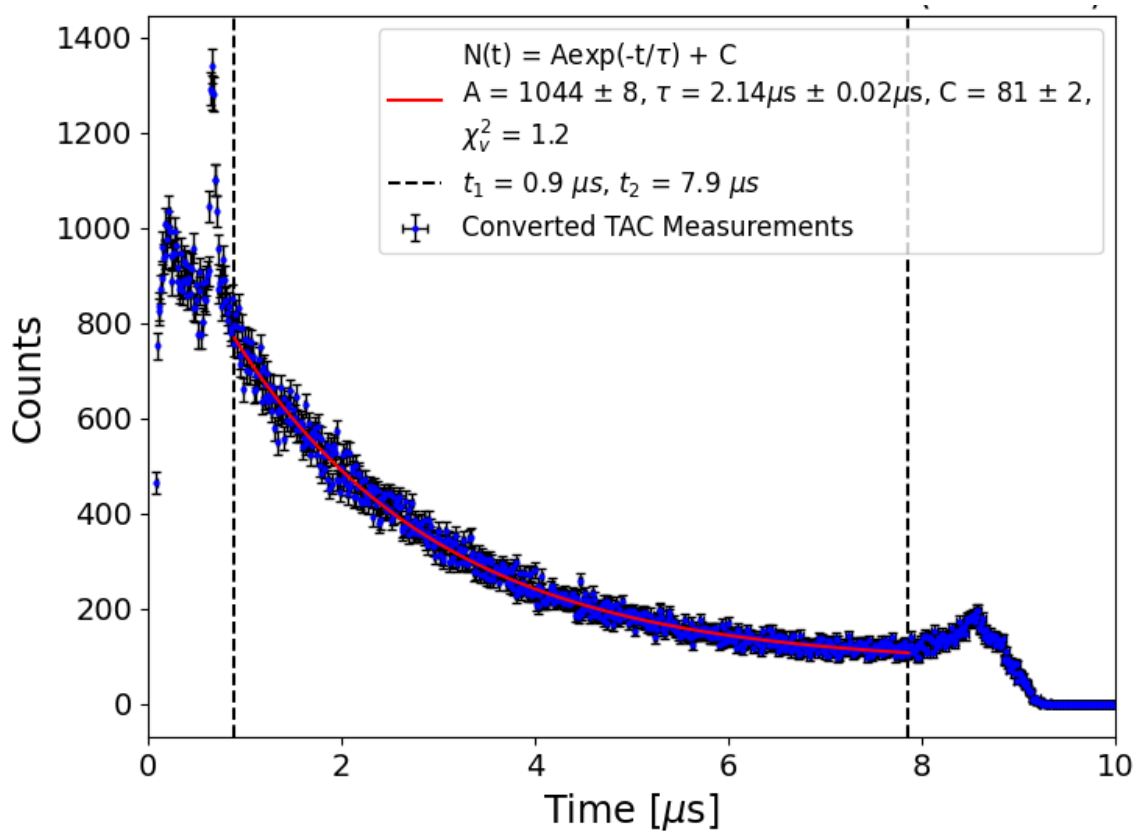


Figure 8: Data obtained from an MCA for the muon lifetime measurement, where a scintillator and PMT setup is used. The voltage measurements from the TAC were converted into time using Eq. 12. Plotted are the counts (y-axis) versus time (x-axis, μs) for a 2-week long measurement, with 1000 bins and a voltage range between 0 to 8V. An exponential decay curve (Eq. 15) was fitted between 0.9μs to 7.9μs and overlayed on top of the data. Error bars in the x and y directions are also included.

The resulting fitting parameters are shown below in Table 4. The mean muon lifetime was extracted to be 2.14 ± 0.02 μs.

Table 4: Results of curve fitting for the muon lifetime measurement, following an exponential decay formula (Eq. 11).

Parameter	Value	Error (±)
A	1044	8
τ	2.14 μs	0.02 μs
C	81	2
χ^2_v	1.2	N/A

The standardized residuals r were also calculated, given by Eq. 16 below

$$r = \frac{N - \widehat{N}}{\sigma_N} \tag{16}$$

where N is the measured counts, and $\widehat{N}$ is the expected counts given by the fitted curve. This is plotted below in Figure 9.

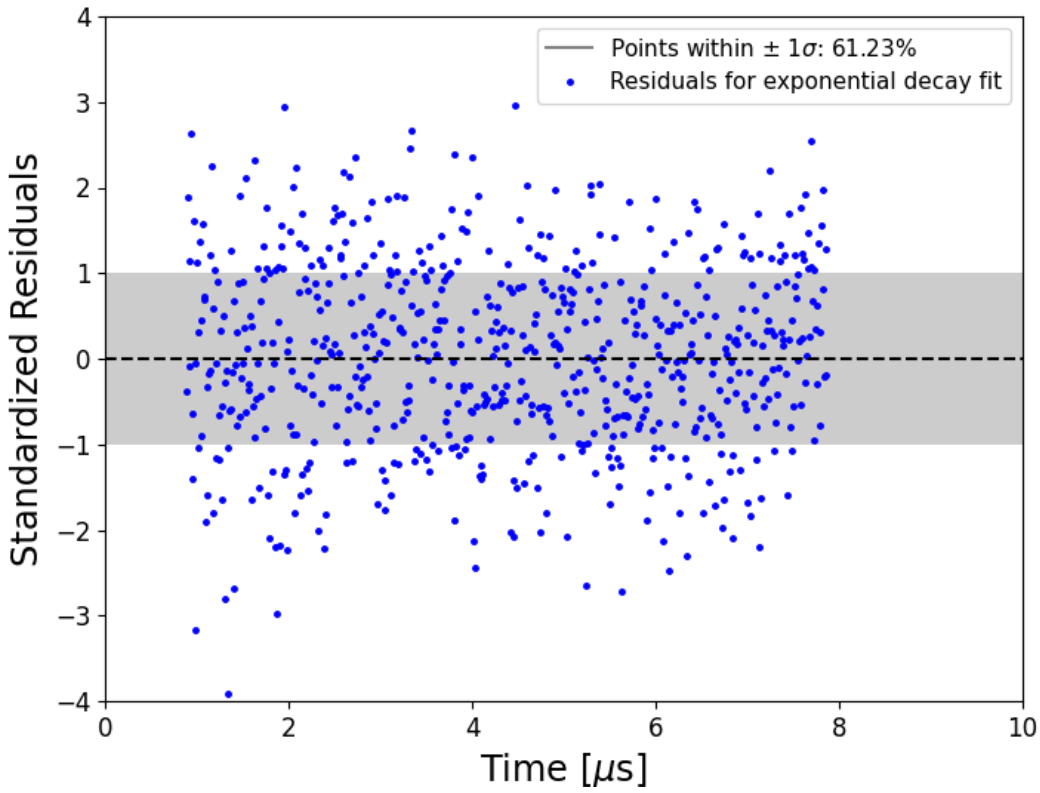


Figure 9: Standardized residuals for the muon lifetime measurement, where a scintillator and PMT setup is used. The voltage measurements from the TAC were converted into time using Eq. 12. Plotted are the residuals (y-axis) versus time (μs) in the region of interest (0.9μs to 7.9μs). The difference was taken between the fitted exponential decay curve and the experimental data and then divided by the standard deviations of each point.

Using the standardized residuals, 61.23%, 99.69%, and 100% of the data lie within 1σ, 3σ, and 5σ, respectively. This indicates the exponential decay model is a good fit to the data and is also statistically significant, consistent with the standard 5σ criterion in particle physics [23].

The event rate measured by the Ratemeter, with the CFD threshold set to 900 mV, was approximately 3,000 min^{-1}. This exceeds the expected rate of ~1340 min^{-1} calculated in the 3.1 Apparatus section. The discrepancy is too large to be solely attributed to background noise and is likely due to the simplified geometric model used for the flux estimate. The predicted rate of coincident events was ~20 min^{-1}, which is consistent with the measured coincident rate. For instance, over a 5-day run, the measured rate R_{meas} was (Eq. 17):

$$R_{meas} = \frac{counts}{time} = \frac{132{,}577}{425{,}934\ \mathrm{s}} = 18.7\ \mathrm{min}^{-1} \tag{17}$$

4.3 Positive and Negative Muon Lifetime Measurement

Since the detected muon decay consists of both positively charged muons (μ^+), which decay freely, and negative charge muons (μ^-), which may undergo nuclear capture in addition to decay, the total decay spectrum is expected to fit a double-exponential decay model [6]. Thus, the collected data from the 2-week long detections of muon decays plotted above in Figure 7 was fit to a model of the form (Eq. 18):

$$N(t) = Ae^{-\frac{t}{\tau_1}} + Be^{-\frac{t}{\tau_2}} + C \tag{18}$$

where τ_1 and τ_2 are the effective lifetimes of the two muon populations, A and B are their respective amplitudes, and C accounts for a constant background.

As in the single-exponential analysis, the data was truncated before 0.9 µs and after 7.9 µs to exclude non-physical regions discussed in Section 5. The resulting fit is shown below in Figure 10.

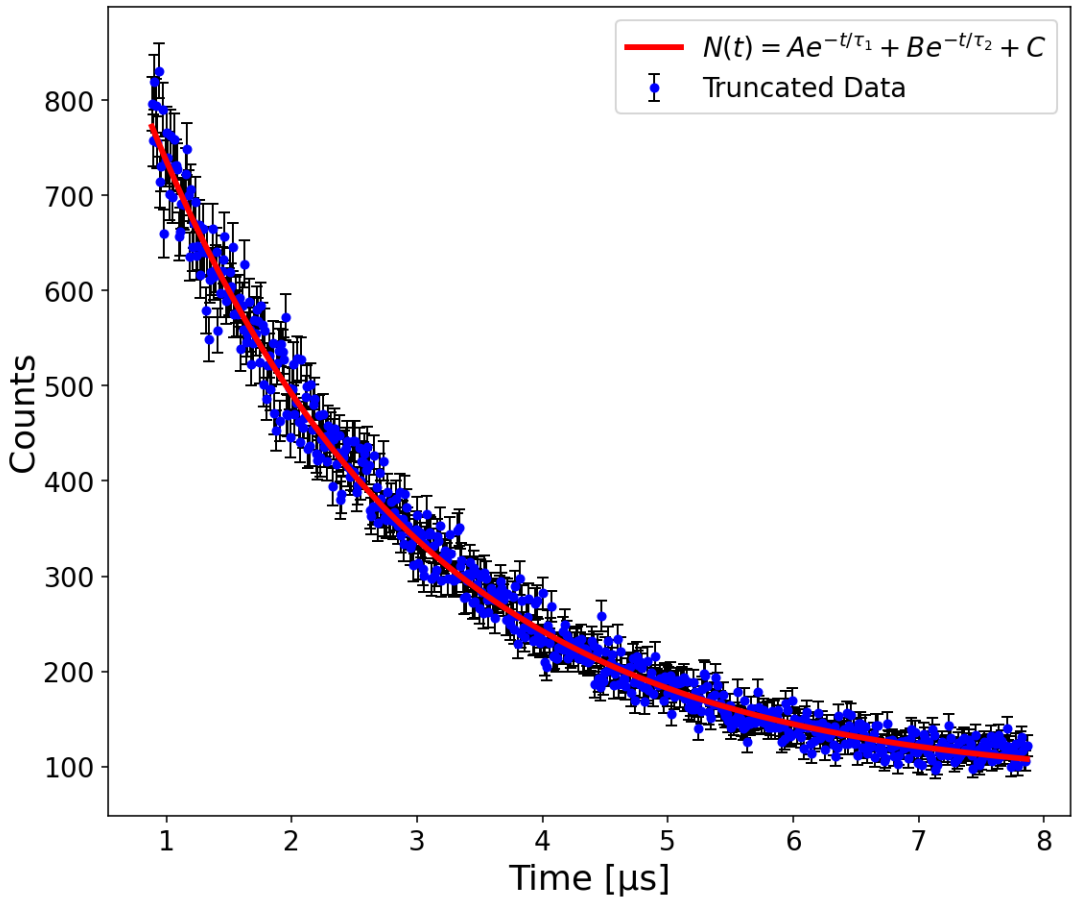


Figure 10: A double-exponential decay curve (Eq. 18) fitted between 0.9µs to 7.9µs and overlaid on top of the 2-week long MCA data. Error bars in the x and y directions are also included.

The resulting fitting parameters for Eq. 18 are shown below in Table 5.

Table 5: Results of curve fitting for the muon lifetime measurement, following a double-exponential decay formula (Eq. 18).

Parameter	Value	Error (±)
A	524	2×10^8
B	522	2×10^8
τ_1	2.14 µs	300 µs
τ_2	2.14 µs	300 µs
C	82	6
χ_v^2	1.2	N/A

The reduced chi-squared value of 1.2 indicates that the model provides an acceptable overall fit for the data. However, the fitted uncertainties on both the amplitudes, A and B, and decay constants, τ_1 and τ_2, were found to be extremely large, several orders of magnitude higher than their respective value. This indicates that the parameters are poorly constrained by the dataset. To investigate the origin of these large parameter uncertainties, the χ_v^2 value was evaluated over a grid of τ_1 and τ_2 values ranging from 1.9 µs to 2.4 µs and plotted in Figure 11.

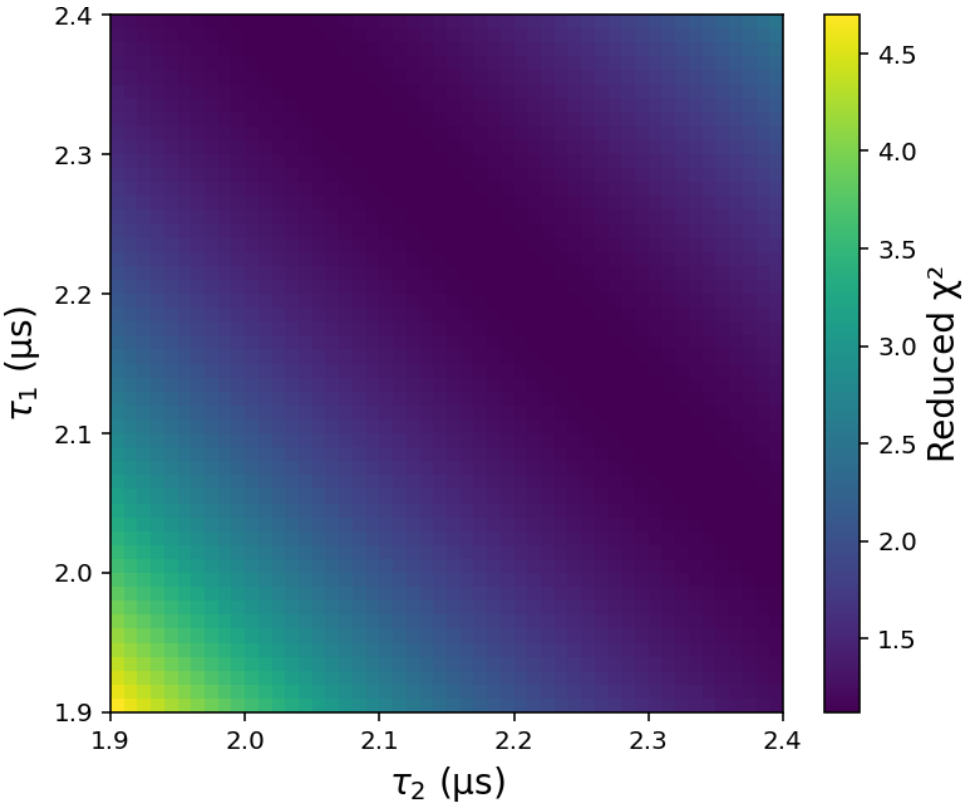


Figure 11: Heatmap of the reduced chi-squared value, χ_v^2, evaluated over a range of decays constants τ_1 and τ_2 (1.9 to 2.4 µs) for the double-exponential muon decay model. Dark blue indicates a χ_v^2 closer to 1, and thus a better fit.

The resulting distribution exhibits a broad, shallow minimum rather than a single well-defined minimum. Decreases in one decay constant can be compensated by increases in the other, provided the amplitudes, A and B, adjust accordingly. This demonstrates a strong degeneracy between τ_1 and τ_2 such that many combinations produce nearly indistinguishable total decay curves.

A second heatmap, Figure 12, was generated by varying the initial input values of τ_1 and τ_2 supplied to the fitting algorithm.

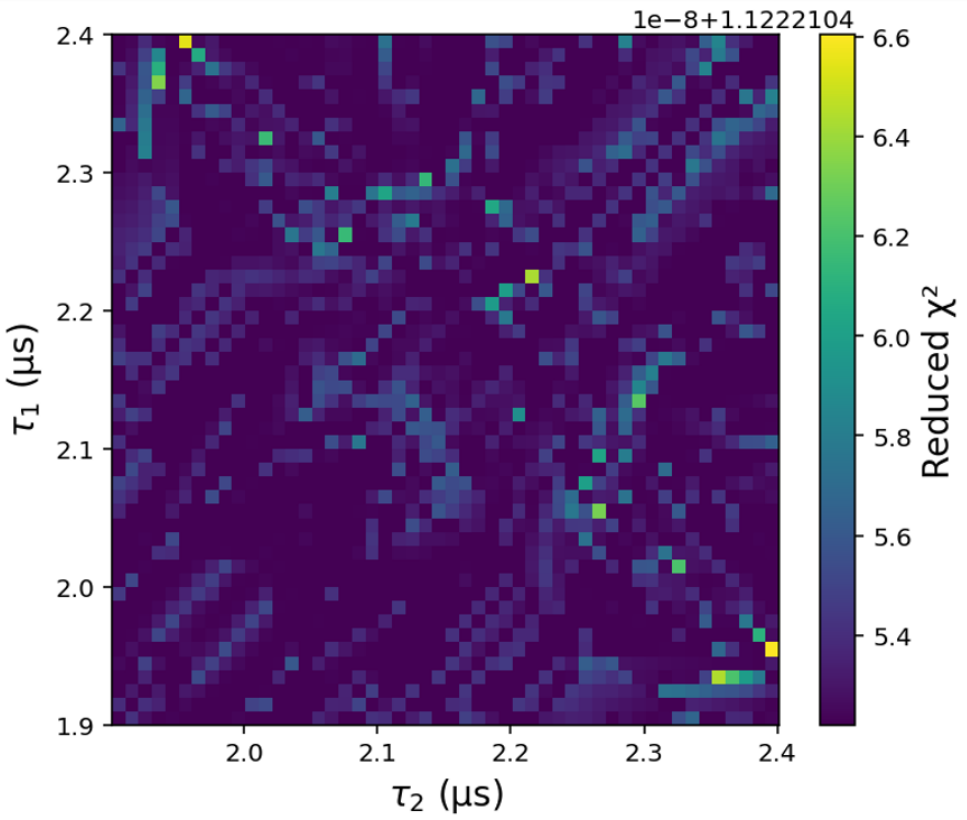


Figure 12: Heatmap of the reduced chi-squared value, χ_v^2, as a function of the initial input values of τ_1 and τ_2 to the curve-fitting algorithm (1.9 to 2.4 μs). The grid has minimal variation in χ_v^2, shown by the scale of the colour bar (1.1222104 + 1×10⁻⁸).

The resulting χ_v^2 values are dominated by numerical noise and differ only at the eighth decimal place. This confirms that the fitting routine does not strongly favour any specific pair of decay constants and that the data lacks sufficient statistical power to resolve two independent lifetimes for positive and negative muons.

To determine if physically motivated constraints on the double-exponential fit improve parameter estimation, additional fits were performed with certain parameters fixed. First, the amplitudes were constrained such that $A = 1.278B$ to match the expected flux ratio [11]. In a second trial, the negative muon lifetime was further fixed at the expected value of $\tau_1 = 2.02\ \mu s$

calculated in the 2 Theory section. The resulting fit parameters for these trials are summarized in Table 6 below.

Table 6: Summary of double-exponential fit results obtained under different parameters constraints. Trial 1 enforces relative amplitudes, while trial 2 additionally fixes the negative muon lifetime of 2.02 μs in the intent of observing a more accurate positive muon lifetime value.

Trial	Fixed Parameter(s)	Curve Fitting Parameter	Value	Uncertainty (±)
1	$A = 1.278B$	A	460	10
		τ_1	2.14 μs	500 μs
		τ_2	2.14 μs	400 μs
		C	82	6
2	$A = 1.278B$	A	459	4
		τ_2	2.23 μs	0.05 μs
	$\tau_1 = 2.02\ \mu s$	C	81	2

Containing only the amplitudes does not significantly reduce the uncertainty in the decay constants, reflecting the strong parameter degeneracy discussed previously. However, fixing the negative muon lifetime yields a well-constrained value for the positive decay constant, $\tau_2 = 2.23 \pm 0.05$ μs.

5 Results and Discussion

This section interprets the results of the muon lifetime experiment and evaluates the suitability of the fitting models used. The single and double exponential fits are discussed, along with limitations imposed by statistical uncertainty and systematic errors identified in the measurement. The implications of these results for extracting physical muon lifetimes are also considered.

5.1 Validity of the Single- and Double- Exponential Models

The muon decay spectrum was analyzed using both single- and double- exponential models to assess whether the data supports the presence of multiple decay constants.

The single-exponential fit yields a well-constrained lifetime of $\tau = 2.14 \pm 0.02$ μs, with a reduced chi-squared value close to unity, and standardized residuals consistent with statistical fluctuations. The measured value is also within uncertainty of the expected mean muon lifetime, $\tau_{mean} = 2.121 \pm 0.008$ μs, supporting the reliability of the measurement and the validity of the single-exponential model.

In contrast, while a double-exponential model is physically motivated by the presence of both positive and negative muons, the fit exhibits strong parameter degeneracy. This is demonstrated by the reduced chi-squared mapping; a wide range of τ_1 and τ_2 combinations provide statistically equivalent fits, resulting in large uncertainties on the fitted parameters.

When physically motivated constraints are applied, enforcing an amplitude ratio and fixing the negative muon lifetime to its expected value, the remaining parameters become significantly better constrained. However, this improvement arises from imposed assumptions rather than additional statistical sensitivity in the data. The resulting positive muon lifetime of $\tau_2 = 2.23 \pm 0.05$ μs is within uncertainty of the expected value of 2.1969811 ± 0.0000022 μs [3]. Although these results agree with expectations, they do not provide definitive evidence that the independent muon lifetimes are distinguishable since the model is heavily constrained.

Overall, the collected data lacks sufficient statistical sensitivity to support the double-exponential model. The single-exponential model thus provides the most reliable description.

5.2 Systematic Error

In the full collected spectrum of muon decays, two sources of systematic error were identified: a sharp peak at ~0.7 μs and a broad deviation at ~9 μs, shown in Figure 13.

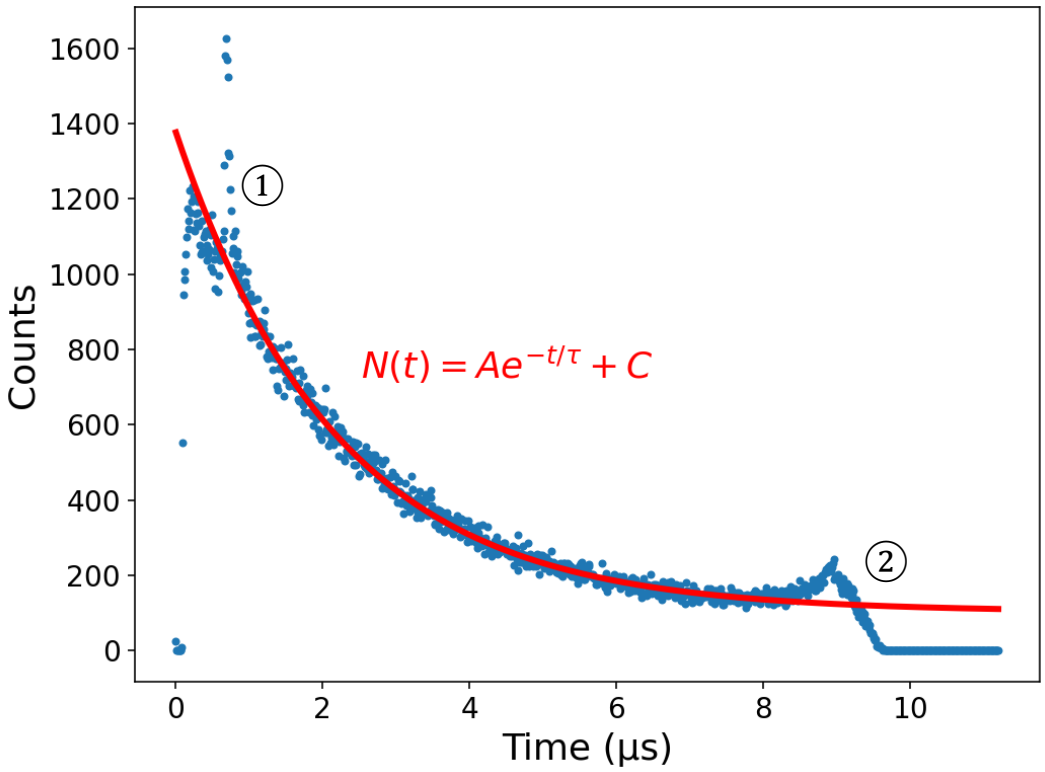


Figure 13: Full muon decay spectrum. Two main systematic features: (1) sharp peak at 0.7 μs and (2) broad deviation near 9 μs.

These features were excluded from the fitting range, and the following discussion describes their sources and likely origins.

5.2.1 Early-Time Peak

A sharp excess of counts is present at approximately 0.7 μs. To test whether the peak was related to the triggering window set by the Gate and Delay Generator, which was originally around 0.8 μs, additional data was collected using a narrower 0.4 μs gate. Both datasets are shown in Figure 14 and the peaks of each are plotted.

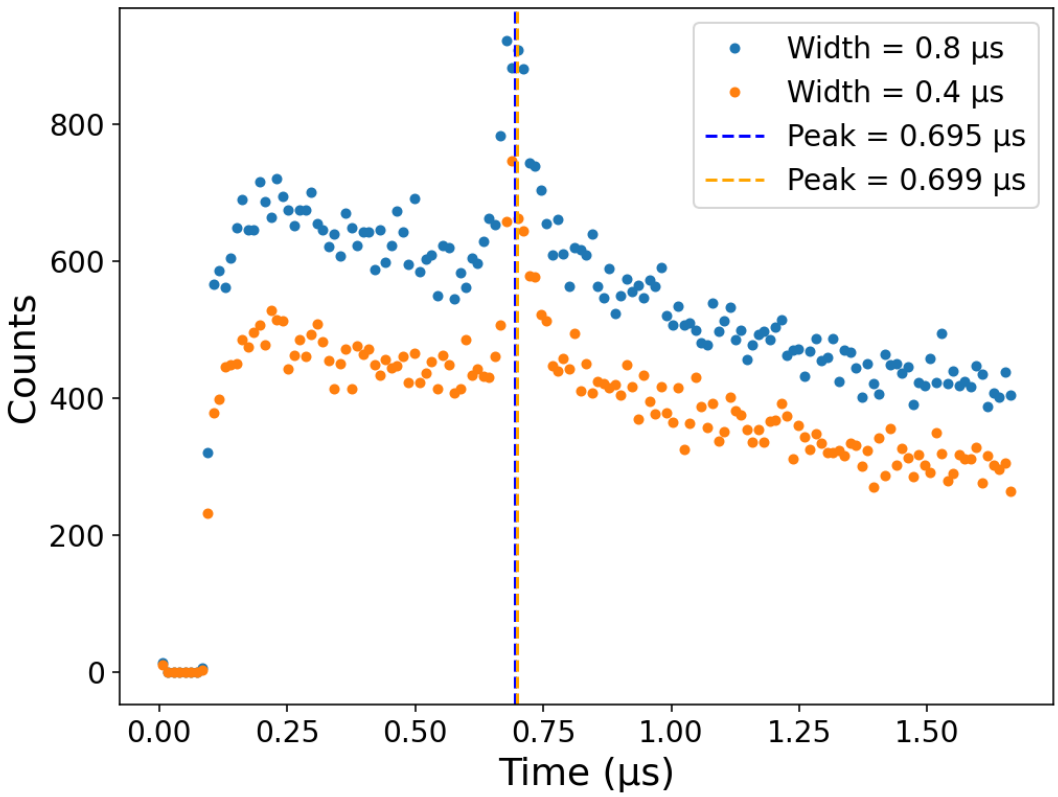


Figure 14: Magnified view of the measured muon decay spectrum around the sharp peak for data collected with 800 ns (blue) and 400 ns (orange) gate widths. Both sets of data are curve-fitted with a Gaussian to determine the peak locations: 0.695 µs and 0.699 µs, respectively.

The peak is observed to remain at the same location in both cases, indicating it is not caused by the gate width. PMT after-pulses can occur at very short delays (tens of nanoseconds) due to electron scattering and, with ion feedback, at delays of several hundred nanoseconds relative to the main pulse [24]. However, the observed feature at 0.7 µs is unusually narrow and sharply defined compared to the broader, continuous distributions typically seen for after-pulses over these ranges.

Possible electronic origins include timing artifacts in the discriminator, reflections or ringing in the signal path, or interactions involving the TAC and Gate and Delay Generator, though further targeted testing would be required to isolate the source [25]. Thus, it is reasonable to conclude that the feature is an electronic artifact, and truncating this early-time region from the exponential fit is justified.

5.2.2 Late-Time Nonlinearity

A broad peak is observed in the muon decay spectrum at ~9 µs, followed by a decay and sharp cutoff at the TAC's maximum range of 10 µs. The TAC's output voltage is given by Eq. 10

$$V = \frac{I\Delta t}{C} \tag{10}$$

where I is the charging current, Δt is the time between start and stop pulses, and C is the capacitance. Near the maximum voltage V_{max}, the effective current decreases with increasing voltage, $\frac{dI}{dV} < 0$, causing long time intervals to be compressed into a narrower voltage range [26]. This is a differential and integral nonlinearity which causes the broad deviation observed in the spectrum near t_{max}.

Integrating the counts above and below the fitted exponential curve, pictured in Figure 15, shows that the excess and deficit largely cancel, resulting in a relative net integral of only ~2%.

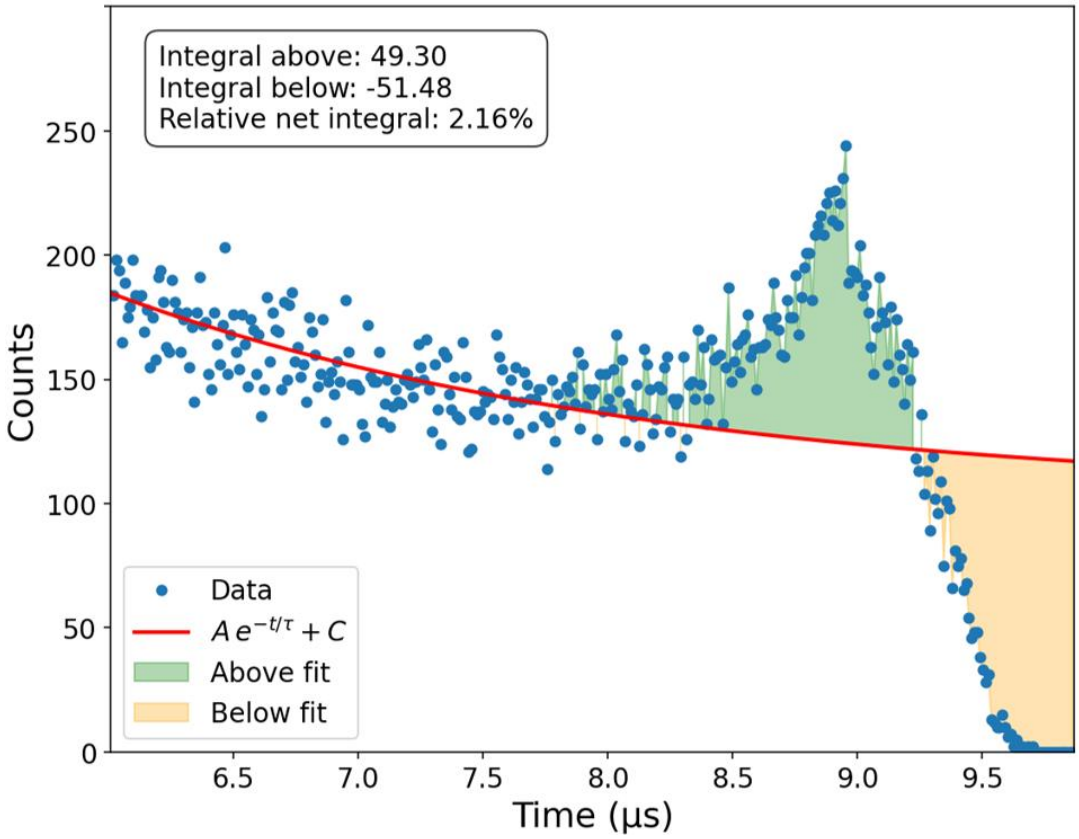


Figure 15: Integrated counts above (green shaded region) and below (yellow shaded region) the fitted exponential curve.

This indicates that the feature corresponds to a redistribution of events rather than the creation of additional decays, consistent with non-ideal TAC effects. Based on this analysis, truncating the late-time region from the exponential fits is justified. Increasing the TAC's range would only shift the onset of this nonlinearity to later times rather than eliminating it, so future muon lifetime experiments should account for this effect when designing the measurement setup.

6 Conclusion

The purpose of this experiment was to investigate fundamental nuclear and particle physics while establishing an experimental value of the mean muon lifetime. This was achieved by measuring coincident events within a plastic scintillator coupled to a photomultiplier tube, and by using Nuclear Instrumentation Modules to isolate muon decay events and measure the time interval between particle entry and subsequent decay.

By fitting 2 weeks of collected data to an exponential decay model, we obtained a value of $\tau_{meas} = 2.14 \pm 0.02$ μs, which is within uncertainty of the theoretically calculated value of 2.121 ± 0.008 μs. This agreement confirms the successful detection and measurement of cosmic ray muon decays, providing experimental validation of relativistic time dilation. Additionally, a double exponential decay model was fitted with constrained parameters, yielding a positive lifetime of $\tau_{\mu^+} = 2.22 \pm 0.04$ μs, which is within error of the accepted value of 2.1969811 ± 0.0000022 μs [3]. However, the data does not have sufficient statistical sensitivity to resolve the lifetimes of positive and negative muons without imposing these constraints.

These results highlight the importance of addressing systematic errors, such as nonlinear TAC behaviour and early-time electronic artifacts, in future experiments. Collecting a larger dataset and optimizing the measurement setup, including timing electronics and the TAC range, would allow for precise determination of positive and negative muon lifetimes.

Appendix

A1 TAC Calibration Data

Table 7: Data used in TAC calibration analysis derived from raw measurements.

Time Interval [μs]	Peak Voltage [V]	Counts	FWHM [V]
0.080000 ± 0.000004	0.0590 ± 0.0006	498	0.034
0.160000 ± 0.000008	0.1190 ± 0.0005	465	0.024
0.32000 ± 0.00002	0.2410 ± 0.0002	499	0.010
0.64000 ± 0.00003	0.47900 ± 0.00004	502	0.002
1.28000 ± 0.00006	0.945 ± 0.001	514	0.054
2.5600 ± 0.0001	1.8690 ± 0.0009	369	0.042

A2 Raw Data Files



Raw_Data_E1.zip

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